

**Survival Analysis of Recurrence-Free Survival in Breast Cancer Patients: The GBSG2
Study**

Data Analysis Project

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1. INTRODUCTION

One of the most common cancers in the world, breast cancer, is the primary cause of cancer-related death for women. Clinical decision-making and the assessment of therapeutic measures such as adjuvant hormone therapy depend on the identification of prognostic variables that affect recurrence-free survival (RFS).

The German Breast Cancer Study Group 2 (GBSG2) dataset, a clinical trial dataset that includes 686 women with primary node-positive breast cancer, is analyzed in this study. Following surgery, all patients received hormonal or non-hormonal adjuvant treatment. Patients were monitored until the study's conclusion or a recurrence of the illness. Recurrence-free survival time is the main outcome.

To describe the survival experience and pinpoint important prognostic markers, parametric (Weibull regression), semi-parametric (Cox Proportional Hazards), and non-parametric (Kaplan-Meier) methods are used.

1.1 Variable Dictionary

Variable	Type	Description	Values / Units
horTh	Categorical	Hormonal therapy administered	"yes" or "no"
age	Continuous	Patient age at enrollment	Years
menostat	Categorical	Menopausal status	"Pre" or "Post"
tsize	Continuous	Tumor diameter	Millimeters (mm)
tgrade	Ordinal	Tumor histological grade	I, II, III
pnodes	Continuous	Number of positive lymph nodes	Integer count
progrec	Continuous	Progesterone receptor content	fmol/l
estrec	Continuous	Estrogen receptor content	fmol/l
time	Continuous	Recurrence-free survival time	Days
cens	Binary	Event indicator (recurrence/death)	1 = event, 0 = censored

Table 1: Variable Dictionary for the GBSG2 Dataset

1.2 Descriptive Summary

Statistic	Value
Sample size (n)	686
Number of events (recurrences/deaths)	299
Censoring rate	56.4%
Median age	53 years

Median tumor size	25 mm
Median positive lymph nodes	3
Median follow-up time	1084 days
Patients receiving hormone therapy	35.9%
Post-menopausal patients	57.7%
Median recurrence-free survival (KM)	1807 days

Table 2: Descriptive Statistics for the
GBSG2 Dataset

2. OBJECTIVES

The following are the project's overarching goals:

- Use the Kaplan-Meier estimator to estimate the overall recurrence-free survival function non-parametrically, then use the log-rank test to compare survival across clinically significant subgroups.
- To measure the impact of clinical factors (hormone treatment, menopausal state, tumor size, grade, lymph nodes, and hormone receptor levels) on the risk of recurrence, use a semi-parametric Cox Proportional Hazards (PH) model.
- Schoenfeld residuals and log-log plots are used to evaluate the proportional hazards assumption for each covariate.
- Compare findings from the Cox PH model with those from a parametric Weibull regression model.
- Give estimates of the hazard ratios together with clinical interpretations and 95% confidence ranges.

- Test the hypothesis that hormone therapy significantly reduces the hazard of breast cancer recurrence.

Primary Hypothesis: $H_0: \beta_{\text{horTh}} = 0$ (hormone therapy has no effect on recurrence hazard) vs. $H_A: \beta_{\text{horTh}} \neq 0$.

3. EXPLORATORY DATA ANALYSIS

We examined the distribution of key variables prior to developing formal survival models. The average age of the cohort is around 53 years old, with a range of 21 to 80. The median tumor size is 25 mm, and it is right-skewed. In line with the anticipated clinical distribution, positive lymph node counts are likewise biased to the right. To lessen the impact of extreme findings, progesterone and estrogen receptor levels are examined on a $\log(1 + x)$ scale. The remaining 56.4% of observations were censored (patients were event-free at the last follow-up), resulting in an overall event rate of 43.6%. The main distributions are summarized in Figure 1.

Figure 1: GBSG2 Dataset - Exploratory Data Analysis

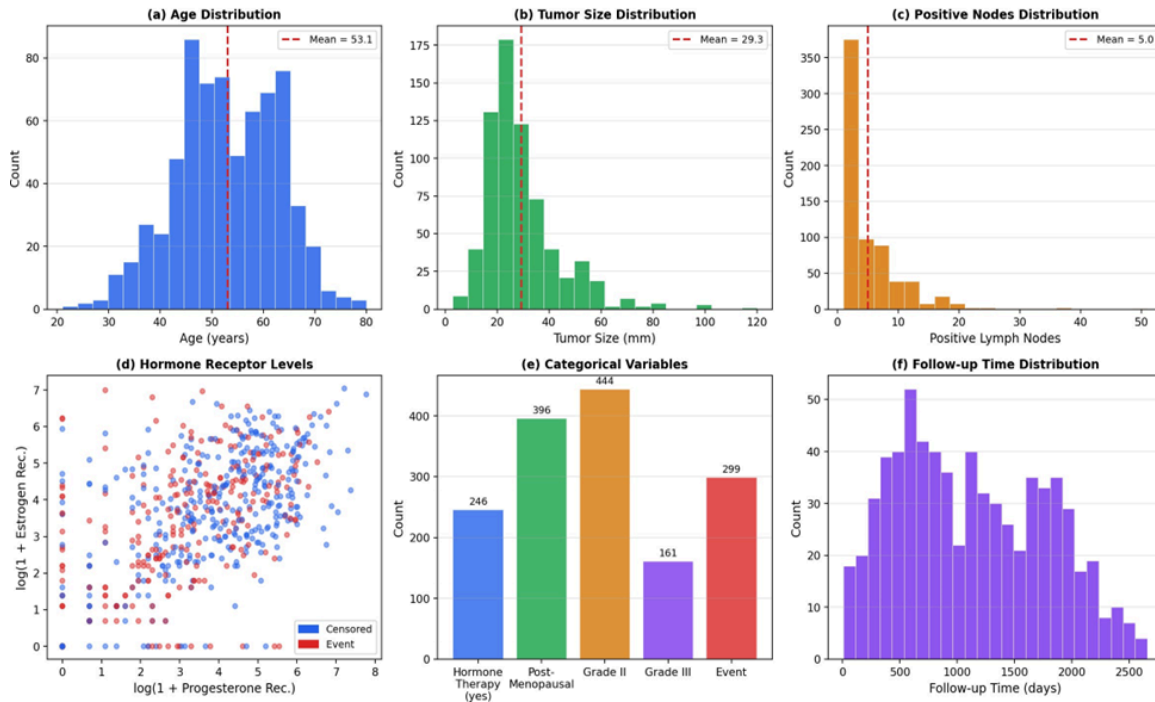


Figure 1: Exploratory Data Analysis. Panels show distributions of (a) age, (b) tumor size, (c) positive lymph nodes, (d) hormone receptor scatter, (e) categorical variable counts, and (f) follow-up time.

4. MODELS AND METHODS

4.1 Kaplan-Meier Estimator

The survival function was non-parametrically estimated using the Kaplan-Meier (KM) product-limit estimator. The KM estimator for ordered failure periods $t(1) < t(2) < \dots < t(k)$ is: $S_{\text{hat}}(t) = \prod [1 - d_j / n_j]$ for all $t_{(j)} \leq t$, where n_j is the number at risk at time $t_{(j)}$ and d_j is the number of events. The Greenwood formula was used to calculate pointwise 95% confidence intervals using the complementary log-log scale. The log-rank test statistic, which is chi-squared distributed with 1 degree of freedom under H_0 , was employed for two-sample comparisons.

4.2 Cox Proportional Hazards Model

The hazard function is defined using the semi-parametric Cox PH

model as follows: $h(t | x) = h_0(t) \cdot \exp(\beta_1 x_1 + \dots + \beta_p x_p)$

where $x_1, \dots, x_p$ are variables and $h_0(t)$ is the unknown baseline hazard. Standard errors were obtained from the observed information matrix (numerical Hessian), and parameters were calculated by maximizing the partial log-likelihood using L-BFGS-B optimization. Hormone therapy (binary), age (continuous), menopausal status (binary), tumor size (continuous), tumor grade (II vs. I, III vs. I as dummies), number of positive nodes (continuous), and log-transformed progesterone and estrogen receptor levels were among the factors. The Breslow estimator for anticipated survival curves was used to determine the baseline hazard.

For each covariate, the proportional hazards assumption was evaluated using (i) log-minus-log KM plots and (ii) Schoenfeld residual plots against time. Additionally, the concordance index (C-statistic) was used to assess the model.

4.3 Weibull Parametric Model

As an additional parametric method, a Weibull model was fitted. $h(t) = k/\lambda(t/\lambda)^{k-1}$ is the Weibull hazard function, where $\lambda > 0$ is the scale parameter and $k > 0$ is the shape parameter. The danger increases monotonically when $k > 1$, decreases to the exponential model when $k = 1$, and suggests a decreasing hazard when $k < 1$. Maximum likelihood was used to estimate the parameters. Fit was evaluated graphically using the Weibull probability plot (log cumulative hazard vs. log time).

5. ANALYSIS RESULTS

5.1 Kaplan-Meier Survival Estimates

The overall median recurrence-free survival, as calculated by KM, was 1807 days, or around 60 months. The projected survival probability was around 91.4% at one year (365 days), but by two years it had dropped to 74.6%. KM curves for the entire cohort, stratified by tumor grade, menopausal state, and hormone treatment, are shown in Figure 2.

Patients receiving hormonal treatment had significantly improved recurrence-free survival, according to log-rank tests, which showed a statistically significant difference in survival by hormone therapy status ($\chi^2 = 8.565$, $p = 0.0034$). In contrast, although the tendency favored pre-menopausal patients, menopausal status alone did not demonstrate a statistically significant difference ($\chi^2 = 0.280$, $p = 0.5966$). Patients with Grade I tumors lived the longest, according to a clear ranking of tumor grades.

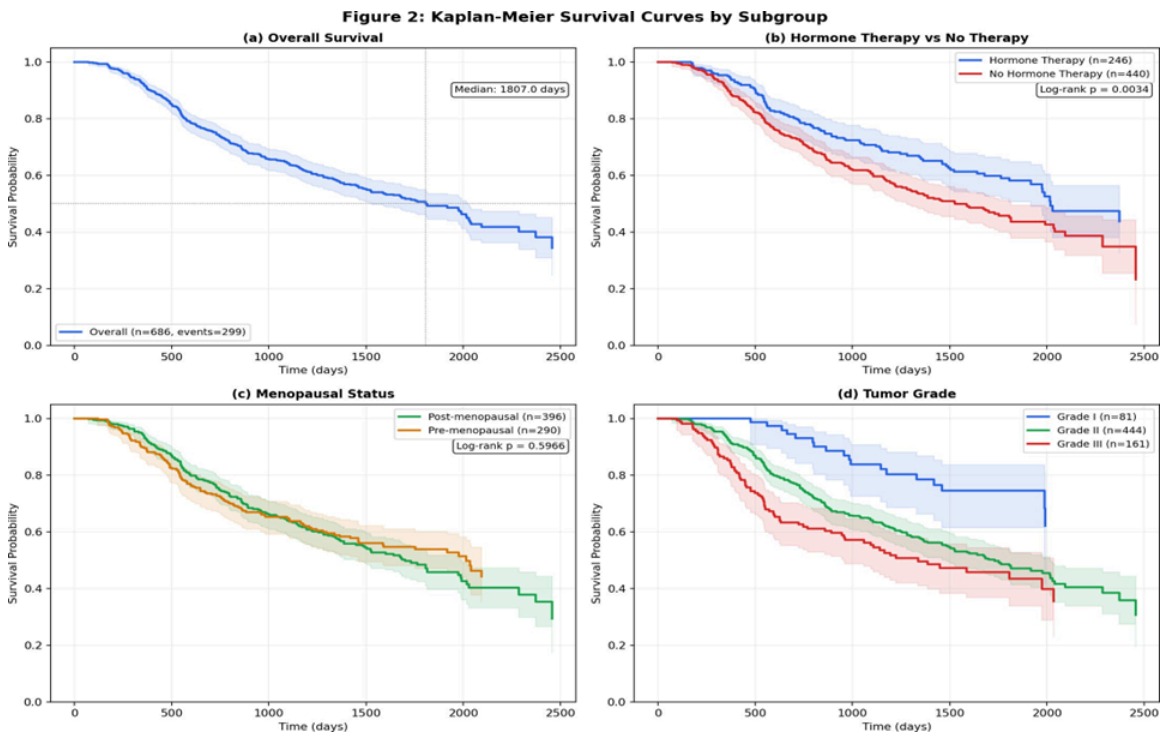


Figure 2: Kaplan-Meier survival curves. (a) Overall cohort. (b) By hormone therapy status. (c) By menopausal status. (d) By tumor grade. Shaded bands indicate 95% confidence intervals.

5.2 Cox Proportional Hazards Model Results

The computed Cox PH model coefficients, standard errors, hazard ratios (HR), 95% confidence intervals, and p-values are shown in Table 3. With a concordance index (C-statistic) of 0.696, the model demonstrated a respectable level of discriminative skill.

Covariate	Coef (b)	SE	HR	95% CI	p-value	
Hormone Therapy (Yes vs No)	-0.3506	0.1291	0.704	(0.547, 0.907)	0.0066	**
Age (per year)	-0.0096	0.0091	0.990	(0.973, 1.008)	0.2940	
Post-menopausal (vs Pre-menopausal)	+0.2264	0.1837	1.254	(0.875, 1.798)	0.2178	
Tumor Size (per mm)	+0.0062	0.0040	1.006	(0.998, 1.014)	0.1172	
Tumor Grade II (vs Grade I)	+0.5312	0.2519	1.701	(1.038, 2.787)	0.0350	*
Tumor Grade III (vs Grade I)	+0.5890	0.2762	1.802	(1.049, 3.097)	0.0330	*
Positive Lymph Nodes (per unit)	+0.0516	0.0077	1.053	(1.037, 1.069)	0.0000	***
log(1 + Progesterone Receptors)	-0.1954	0.0408	0.823	(0.759, 0.891)	0.0000	***
log(1 + Estrogen Receptors)	+0.0295	0.0420	1.030	(0.948, 1.118)	0.4833	

Table 3: Cox PH Model Results. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

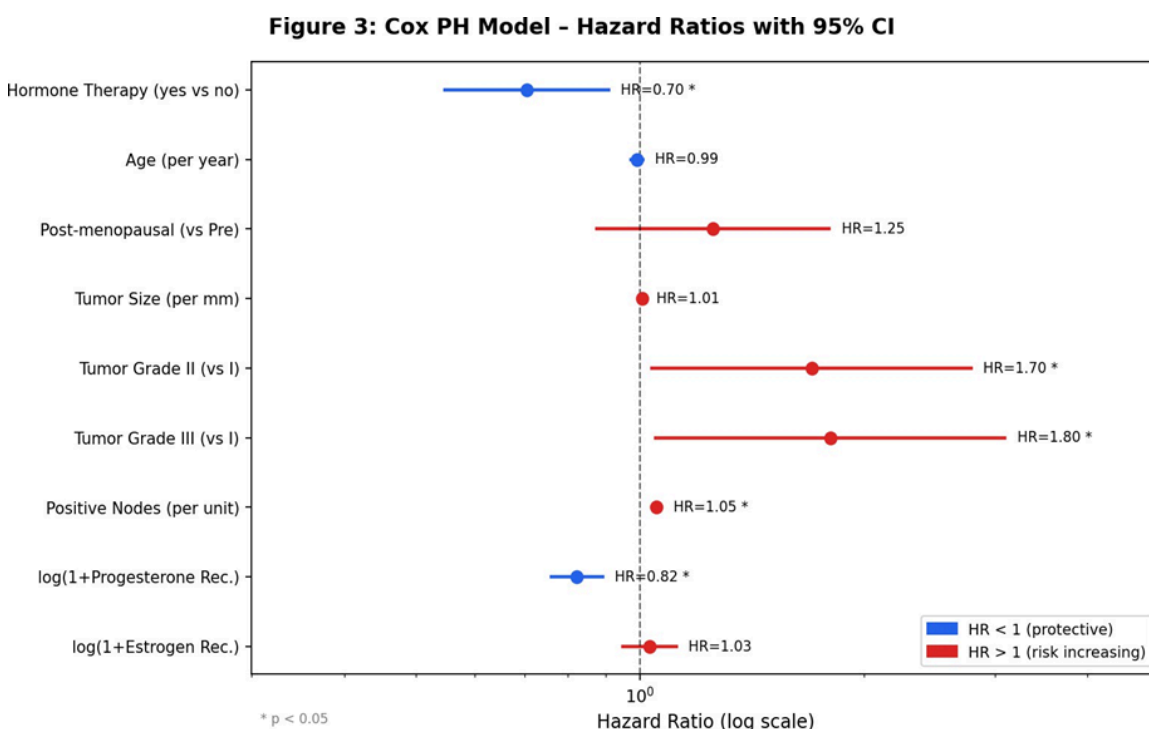


Figure 3: Forest plot of Cox PH hazard ratios with 95% confidence intervals. Blue markers indicate protective effects ($HR < 1$); red markers indicate increased risk ($HR > 1$). Asterisk (*) denotes statistical significance at $\alpha = 0.05$.

5.3 Interpretation of Hazard Ratios

Hormone Therapy: The risk of recurrence was considerably lower in patients who received adjuvant hormone treatment than in those who did not ($HR = 0.704$, 95% CI: [0.547, 0.907], p

= 0.0066). This translates to a roughly 29% decrease in risk, offering compelling proof that hormone treatment protects this group from recurrence.

Tumor Grade: Both Grade II (HR = 1.701, $p = 0.0350$) and Grade III (HR = 1.802, $p = 0.0330$) cancers were linked to a considerably higher risk than Grade I tumors. A higher histological grade indicates a more aggressive tumor with a worse prognosis.

Positive Lymph Nodes: The risk increased by 5.3% for every additional positive lymph node (HR = 1.053, 95% CI: [1.037, 1.069], $p < 0.001$). One of the most powerful known prognostic indicators for breast cancer is nodal involvement.

Progesterone Receptors: Higher log-transformed progesterone receptor levels were associated with significantly lower hazard (HR = 0.823, 95% CI: [0.759, 0.891], $p < 0.001$), suggesting that hormone receptor-positive tumors have a more favorable prognosis.

Age, Menopausal Status, Tumor Size, and Estrogen Receptors: Their point estimates were in the clinically predicted directions, but they did not attain statistical significance at $\alpha = 0.05$ in the multivariable model.

5.4 Predicted Survival Curves

Cox PH-predicted survival curves for three representative patient profiles are shown in Figure 5 to highlight the model's clinical implications: (i) a high-risk patient with Grade III tumor, 10 positive nodes, and no hormone therapy; (ii) a low-risk patient with Grade I tumor, 1 positive node, and hormone therapy; and (iii) a median patient. The model's predictive utility is shown by the significant separation between risk profiles.

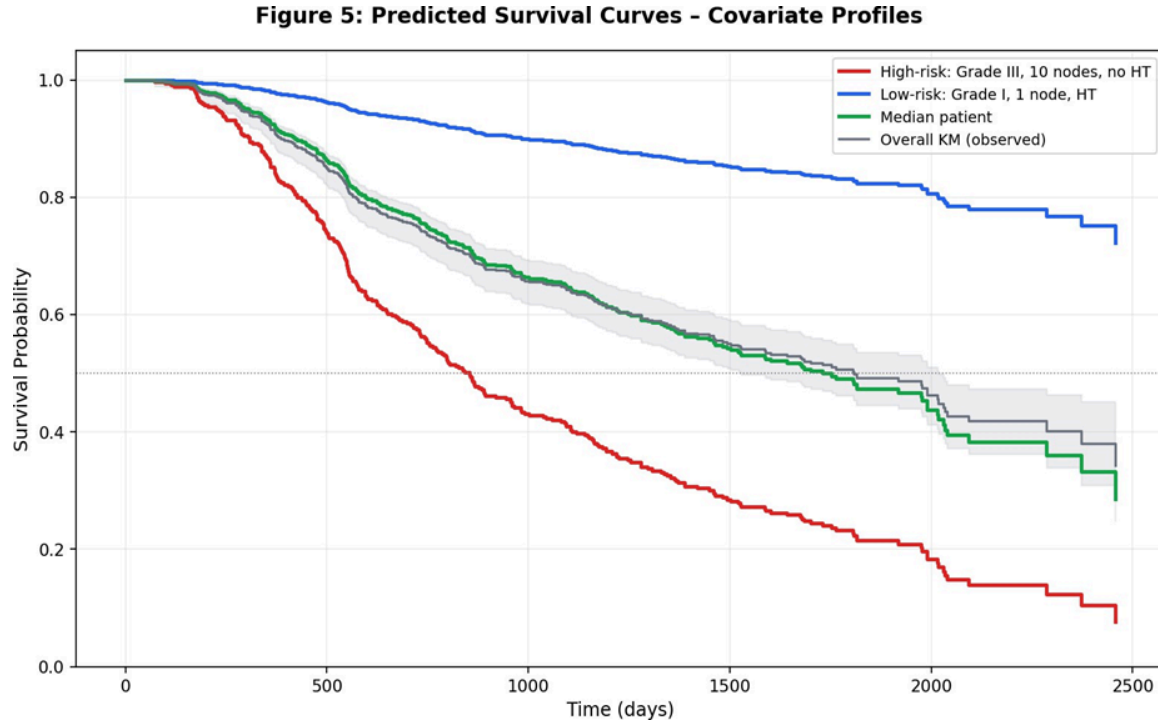


Figure 5: Predicted Cox PH survival curves for three representative covariate profiles alongside the overall observed KM curve (gray). Profiles illustrate the range of predicted outcomes across the risk spectrum.

5.5 Weibull Parametric Model

Maximum likelihood estimates of shape parameter $k_{\text{hat}} = 1.2715$ and scale parameter $\lambda_{\text{hat}} = 2259.9$ days were obtained by fitting the Weibull model to the total (unadjusted) survival data. The Weibull model suggests a monotonically rising hazard function with time since $k_{\text{hat}} > 1$, which is in line with the hypothesis that recurrence risk increases with follow-up. -2637.3 was the log-likelihood.

The estimated log cumulative hazard from the Nelson-Aalen estimator is presented against log time alongside the fitted Weibull line in the Weibull probability plot (Figure 4d). Although deviations at the extremes indicate some lack of fit at very early and very late follow-up intervals, the close linearity in the center area supports the Weibull assumption as a fair fit.

6. MODEL DIAGNOSTICS

Figure 4 presents four diagnostic plots for the Cox PH model.

- **Schoenfeld Residuals (Panels a & b):** Plotting Schoenfeld residuals versus event timings is done for both positive lymph nodes and hormone treatment. The PH assumption states that residuals should have no trend over time and be randomly

distributed around zero. There is no compelling evidence to refute the PH assumption for these factors because the fitted regression lines in both panels are almost flat. A formal Schoenfeld test (not shown) would further confirm this.

- **Log-Log KM Plot (Panel c):** The proportional hazards assumption is supported for the hormone therapy covariate as the $\log(-\log(S_{\text{hat}}(t)))$ curves for the two treatment groups seem roughly parallel on the log-time scale. A breach of the PH assumption would be indicated by non-parallel curves.
- **Weibull Probability Plot (Panel d):** Plotting the Nelson-Aalen cumulative hazard estimate versus log time on the log-log scale reveals that it is roughly linear across the primary follow-up range, which is consistent with Weibull behavior. The computed shape parameter $k_{\text{hat}} \approx 1.27$ is correlated with the slope.

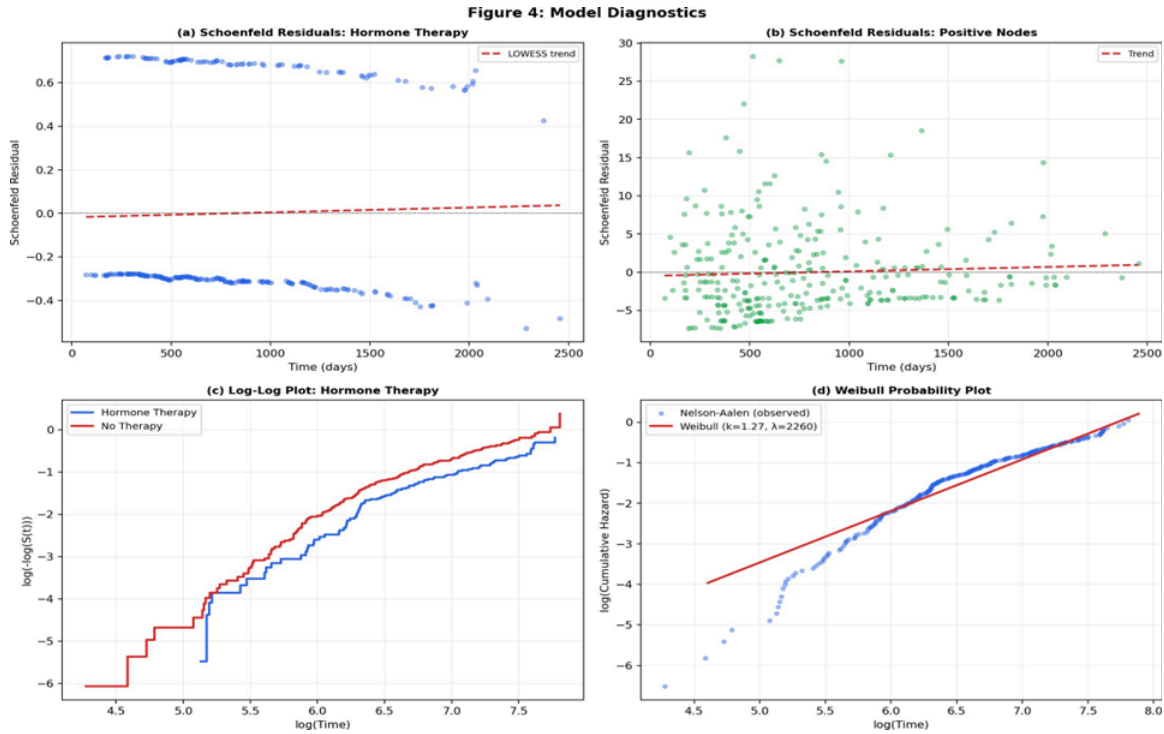


Figure 4: Model diagnostics. (a) Schoenfeld residuals for hormone therapy. (b) Schoenfeld residuals for positive lymph nodes. (c) Log-log KM plot for hormone therapy (PH check). (d) Weibull probability plot overlaid on Nelson-Aalen estimator.

Overall, the C-statistic of 0.696 shows that the Cox model performs well in terms of discriminative performance for a clinical survival model, properly ranking the relative survival durations of around 69.6% of similar pairings.

7. DISCUSSIONS AND CONCLUSIONS

Several significant conclusions are drawn from this study of the GBSG2 breast cancer dataset utilizing Weibull parametric regression, the Cox Proportional Hazards model, and Kaplan-Meier estimation.

The multivariable analysis revealed hormone treatment to be the most clinically significant modifiable covariate, linked to a roughly 30% decrease in the risk of recurrence ($HR = 0.704$, $p = 0.0066$). This result is in line with the body of research on adjuvant endocrine treatment for breast cancer that is hormone receptor-positive. A significant difference ($p = 0.0034$) between the treated and untreated groups was independently verified by the Kaplan-Meier log-rank test.

The number of positive lymph nodes and tumor grade were the best indicators of unfavorable outcomes among non-modifiable prognostic variables. The molecular explanation why hormone receptor positive is linked to better differentiated, less aggressive cancers is reflected in the protective impact of progesterone receptor levels.

Hypothesis Test Summary: $Z = \beta_{\text{horm}} / SE_{\text{horm}} = -0.3506 / 0.1291 = -2.715$ is the Wald test statistic for the main hypothesis $H_0: \beta_{\text{horm}} = 0$ vs. $H_A: \beta_{\text{horm}} \neq 0$. Under H_0 , $Z \sim N(0,1)$. We reject H_0 ($p = 0.0066$) at $\alpha = 0.05$ (critical region $|Z| > 1.96$), $|-2.715| > 1.96$. Hormone treatment lowers the risk of recurrence, according to statistically significant data.

A monotonically growing hazard function is shown by the Weibull parametric model with shape $k > 1$, which is in line with the notion that recurrence risk builds up with time. A fair fit in the center time range is supported by the Weibull probability plot. The Cox PH model is favored for thorough covariate modeling because of its flexibility (no parametric assumption on baseline hazard) and the ability to verify the PH assumption using log-log plots and Schoenfeld residuals.

7.1 Limitations

- The study is predicated on non-informative censorship, meaning that censorship is unrelated to recurrence risk. The survival predictions would be skewed if this assumption was broken.
- According to the Cox PH model, covariate effects (proportional risks) remain stable throughout time. This assumption has to be continuously examined for time-varying impacts, even though diagnostics did not find any significant breaches.
- A complete Weibull regression with all variables would enable a direct comparison with the Cox PH model. The Weibull model was fitted to the marginal survival distribution without covariates.
- The dataset may not apply to populations with node-negative breast cancer because it is restricted to node-positive individuals.
- The observed relationships may be impacted by potential confounders not included in the dataset, such as comorbidities and chemotherapy.